MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R09

Demo navigation guide

Version 1.2, 5 November 2026: the 20-minute demo, click by click

The question this guide answers

What exactly do I click, show and say in each minute of the 20-minute demo, and what do I do when Foundry is slow?

One-line answer

Part A walks the safety officer's morning in the app (0:00 to 7:30). Part H goes under the hood in the deck's slide order (7:30 to 20:00). Every stop has a click path, a screenshot, a key line, the rubric row it proves, a seconds budget and a fallback.

What this is	What this is not
• The presenter's click path for the live demo, in the binding agenda order • A pre-demo checklist: tabs, zoom, what to rebuild (nothing), the triage choice (Cancel by default) • A fallback screenshot for each live stop, and prepared answers to the three rubric-check questions	• The deck or its full spoken words (see `docs/11-handin/deck/speaking_script.md`) • The full ripple map or question bank: `demo/ripple-table.md` and `demo/rehearsal-plan.md` • A claim that any signal is caused by a drug

How to read this. New to the project? Read **R00 Start here** first (`docs/11-handin/reports/pdf/R00-start-here.pdf`). Chapter 1 is the whole demo at a glance. Chapter 2 is the setup, done before anyone joins. Chapters 3 and 4 are the stops: grey labels on the left, what to do on the right; the **Say** row is the key line (full words in the speaking script). Chapter 5 is for when something breaks and for the hardest questions. Numbers come from `numbers.json`, captured in Foundry at the baseline gate OBSERVED; other figures name their record in brackets.

Contents	Page
01 · The 20 minutes at a glance	2
02 · Before the demo	6
03 · Part A: as the safety officer (0:00 to 7:30)	9
04 · Part H: under the hood (7:30 to 20:00)	17
05 · Plan B and hard questions	27

01

The 20 minutes at a glance

The agenda is fixed in the hand-in brief, and the deck, the speaking script and this guide all follow it. The user is the hospital pharmacy safety officer (we call her Dana). She has three morning questions: which drugs have side effects reported that the label does not mention, which have an active recall, and what changed since her last review. Her one rule: every flag must trace back to its source.

The 13 formulary drugs. Opioids (8): oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol. GLP-1s (5): semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide. Exenatide has no reference label, so signals exist for 12 of 13 (A013).

What this means for Dana (the safety officer)

1,073 is a review queue, not 1,073 findings. A signal = reported more often than for other drugs (Evans rule) **and** not found in the drug's reference labels, or only possibly in label (108: found only through a medium-band synonym).

NAION is a known positive, not a discovery. EMA's safety committee (PRAC) concluded on 6 June 2025 that NAION is a very rare side effect of semaglutide (WHO alert 27 June 2025); the 3 US label versions we search still do not mention it (DECISIONS learning 2026-11-05). A monitor that missed it would be broken.

NAION was not checked by the AI. The AI has read 0 semaglutide labels (the pilot is 2 oxycodone labels). The method code `text+ai_v1+syn_v2` names the rule version, not which checks ran.

'Matched to our products' means the 658 FDA-listed products we track (foundry README), not her shelves. We have no inventory data. And she cannot open the 394 reports from her screen yet.

Card 3 counts the last 30 days, not since yesterday. Page 3's date range narrows it to "since my last review".

Only the project owner can submit a triage decision today. In the demo the form is opened and **cancelled** unless the owner decides otherwise.

New signals arrive only with a new FAERS quarter, and FAERS is about 130 days behind (monitoring `faers_age_days`).

Stop	Time	s	Deck	Proves (rubric row)
A0 The officer's problem	0:00–1:30	90	1–2	Communication
A1 Morning brief	1:30–3:00	90	3	Application (trust)
A2 Noise, then semaglutide	3:00–3:40	40	4	Application
A3 Evidence (NAION)	3:40–4:30	50	4	Application (trace)
A4 Labels searched + DailyMed	4:30–5:20	50	4	Application, AI reliability
A5 Triage form (Cancel)	5:20–6:00	40	4	Application (trust)
A6 Recalls + FDA record	6:00–6:50	50	5	Application, Data judgment
A7 What changed	6:50–7:30	40	5	Application
H1 Architecture picture	7:30–8:30	60	6	Architecture
H2 Data Lineage	8:30–9:30	60	6	Architecture
H3 Ontology picture	9:30–10:15	45	7	Architecture
H4 Ontology Manager	10:15–10:45	30	7	Architecture

Stop	Time	s	Deck	Proves (rubric row)
H5 Object Explorer trace	10:45–11:30	45	7	Application (trace)
H6 One report to a signal	11:30–12:30	60	8	Data judgment
H7 Signal code (optional)	12:30–12:50	20	8	Architecture
H8 Review queue, not findings	12:50–13:30	40	9	Data judgment
H9 AI reliability	13:30–14:45	75	10	AI reliability, Optimization
H10 Monitoring	14:45–15:30	45	11	AI reliability, Data judgment
H11 Optimization log	15:30–16:45	75	12	Optimization
H12 Incremental FAERS	16:45–17:30	45	13	Optimization
H13 Data judgment	17:30–19:00	90	14	Data judgment
H14 Another month	19:00–19:40	40	15	Communication
H15 What to take away	19:40–20:00	20	16	Communication
Total		**1,200**		

Staying on time

Stop codes are A0–A7 (the app) and H1–H15 (under the hood), so they never clash with the deck's backup slides B1–B12. Checkpoints: 3:00, 6:00, 7:30, 11:30, 15:30, 19:00. More than 30 s late: drop the optional switches (H7, H9, H10, H12, H13), then the Object Explorer half of H5, then shorten H12 to one sentence. Never cut A4 (labels searched and the DailyMed search), A5 (the Action form), H2 (lineage) or the honest limits in H9.

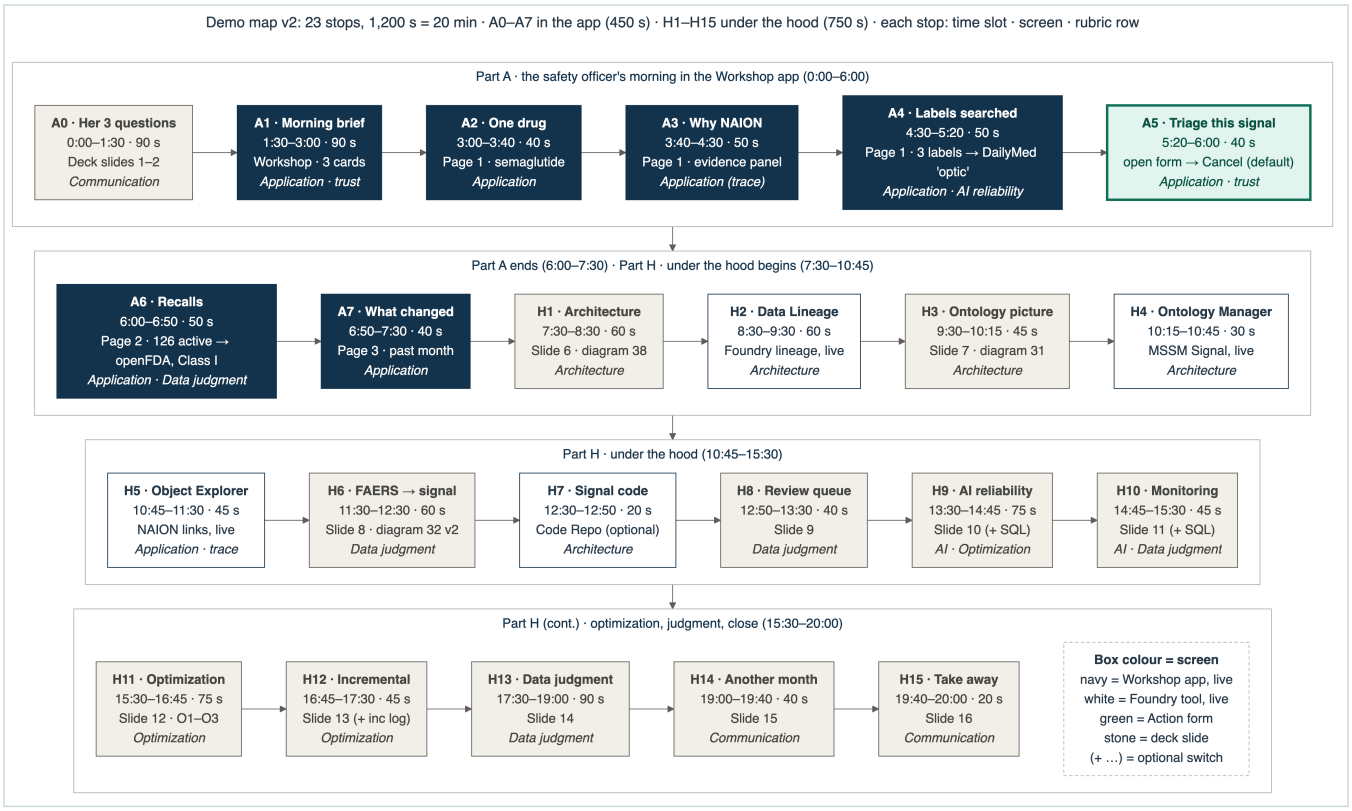


Figure 1. The demo map (diagram 36 v2): the officer's path first, then the stops under the hood. Its slot times and rubric rows match the table above.

Source: docs/diagrams/png/36-handin-demo-map-v2.png

Words to explain the first time you use them

Term	Say
Foundry	Palantir's data platform: pipelines, a data model and apps in one place
Dataset, transform	A table in Foundry; the Python code that builds one table from others
Code Repository	Foundry's built-in code editor, with git and a Build button
Data Lineage	Foundry's live graph of which dataset is built from which
Ontology, object type, link, Action	The business layer on top of datasets: Drug, Signal, Recall as things with links between them. An Action is a form that writes a change back
Workshop	Foundry's app builder; our app is the module MSSM Safety Officer (v0.9.0, 4 pages)
AIP	Foundry's AI platform; we call Claude Sonnet 5.5 through it
FAERS	The FDA's database of adverse event reports, published as quarterly files
PRR	Proportional reporting ratio: how much more often a reaction is reported with this drug than with all other drugs
Evans rule	A signal needs at least 3 reports, PRR at least 2 and chi-square at least 4 (D006, proposed)
MedDRA preferred term (PT)	The standard medical term FAERS uses for a reaction
DailyMed, SPL	The NIH site for official drug labels; the XML format they come in

Term	Say
openFDA	The FDA's public API: recalls (enforcement), Drugs@FDA and NDC
RxNorm, RxCUI	The NIH drug naming standard; its ID for one ingredient
NDC	National Drug Code: the ID printed on a package
Evaluation id	A fingerprint of the evidence behind a signal (report count, PRR, label-check version, FAERS window, labels searched, labels read by AI). It changes when any of those change

02

Before the demo

The day before (then run rehearsal pass 1)

#	Check	Pass when	If not
1	**Do not rebuild the raw pulls.** The demo uses the frozen baseline data	Morning brief shows 1,073 and 126	If someone rebuilt, the lead re-captures numbers.json before the slides change
2	Data Health on `ontology_signal_signals` (Health tab)	Build status and Primary key both Passed	Dataset page: Build, then Force, Only this dataset; check again
3	AI cache rows (SQL on `ai_label_section_extractions`)	9 rows, all `from_cache` true, `n_reactions` = `n_grounded`	Use the screenshot. Do not rebuild the AI transform on demo day
4	**Owner's choice for the triage form: Cancel (default) or Submit**	The owner has said which, before the demo	No answer means Cancel. Submit creates a permanent object: the Action is create-only, with no delete
5	If Submit was chosen: logged in as the **project owner**	Triage form opens	Only the owner can submit today. Otherwise Cancel
6	Keep the 2 test decisions	Still in Object Explorer	See the note below
7	Rehearse the form by **opening it and pressing Cancel**; never Submit in rehearsal	NAION's decisions table is still empty	If a decision landed by mistake: keep it, tell the lead
8	**DailyMed check (A4):** click the Ozempic v20 link on page 1, Cmd+F `optic`	First hit is not the retinopathy warning; no optic neuropathy found. Note the version DailyMed shows	Use `ischemic optic`. Newer version shown: say so. Screenshot the search as the fallback
9	**Class I count (A6):** page 2, Is Active = Yes, click the Class column	Active Class I count and D-0788-2026's product match written down (**not measured** today)	Not counted: say no number
10	**Stopwatch walk-through by a person** of the three questions, reading included	Time written down. 'Under two minutes' is a target until then	Say 'designed for under two minutes, not yet timed with a person'

Why keep the two test decisions

The Action is create-only, so there is no delete Action; removing the test decisions would mean adding one. They are audit evidence that a decision survives a rebuild. Their rationales say STEP 6 ACTION SAVE TEST (recall D-0338-2026) and STEP 8 WORKSHOP TEST (Oxycodone · Myoclonus). Neither is on NAION, so neither appears in Part A. Worth checking in rehearsal (not verified): the Myoclonus decision stores evaluation id cbcd091262d23bd6 from step 8 (A012); the label check has changed twice since, so the signal's current id should differ. If it does, it is a live example of a decision on older evidence.

Thirty minutes before: windows and tabs

Window 1: Chrome with Foundry. Window 2: the deck in presenter view. Window 3 (backup): every image in `docs/11-handin/screens/` open in Preview. Hide the bookmarks bar, turn on Do Not Disturb, close chat and email. Browser zoom 110% for the app and dataset tabs, 125% for the code tabs. Cmd+1 to Cmd+8 jump straight to tabs 1 to 8; Cmd+9 jumps to the last tab (the openFDA record).

Tab	Open	Leave it showing	At
1	Workshop module **MSSM Safety Officer** v0.9.0, View mode (folder `Workshop/`)	Morning brief. Click all 4 tabs once to warm them, then come back	1:30
2	`Data/curated/ontology_signal_signals`, Explore pipeline, Explore data lineage; expand parents 3 levels	The graph laid out as in the H2 screenshot. Do not press Layout again	8:30
3	Ontology Manager, **MSSM Signal**	Overview	10:15
4	Object Explorer, search `Semaglutide · Optic ischaemic neuropathy`	First result selected, Links tab	10:45
5	Repo `Pipelines/pipelines_clean`, file `ontology_signals.py`	Lines 22 to 29 (Foundry numbering); SQL panel with the **monitoring** query already run	12:30, 14:45, 17:30
6	Same repo, second tab, `ai_label_extraction.py`	Lines 16 to 25; SQL panel with the **AI cache** query already run	13:30
7	`Data/curated/ontology_signal_signals`	Health tab	14:45
8	`Data/clean/clean_faers_inc_log` (optional)	Preview	16:45
9	Last tab. The FDA's own record: `https://api.fda.gov/drug/enforcement.json?search=recall_number:"D-0040-2027"` (the openFDA record the pipeline reads)	The JSON: number, class, status, product, lots. Cmd+9 jumps here	6:20

Watch out

The code repository is slow to start ('Starting Python server'). Open tabs 5 and 6 at least 10 minutes early. **Never type in the editor** during the demo: an accidental edit can be committed. The SQL panel is slow, so run the queries during setup and leave the results showing.

```
-- tab 6: AI cache rows (H9)
SELECT label_key, section_name, status, n_reactions, n_grounded, from_cache,
       prompt_tokens, completion_tokens, called_at
FROM `/kestrel-01f4ab/Medication Safety Signal Monitor/Data/curated/ai_label_section_extractions`
ORDER BY label_key, section_name
```

```
-- tab 5: latest monitoring run (H10, H13)
SELECT check_id, area, value, previous, threshold, status, detail
FROM `/kestrel-01f4ab/Medication Safety Signal Monitor/Data/monitoring/monitoring_checks`
WHERE run_at = (SELECT MAX(run_at) FROM
`/kestrel-01f4ab/Medication Safety Signal Monitor/Data/monitoring/monitoring_checks`)
ORDER BY status DESC, area
```

```
-- Q&A only: what the AI output looks like (term, assertion type, grounding, quote)
SELECT section_name, normalized_term, assertion_type, grounding, quote
FROM `/kestrel-01f4ab/Medication Safety Signal Monitor/Data/curated/ontology_labeled_reactions`
WHERE section_name = 'Boxed Warning' ORDER BY label_key, normalized_term LIMIT 12
```

Signal to pre-select: none. Leave page 1 unfiltered, and choose semaglutide live, so the room sees the filter work. NAION is then row 1.

Data freshness: the exact words

"The numbers are frozen from our baseline build on 5 November. FAERS runs through 2026 Q2, the latest quarter the FDA has published; our freshness check puts it at about 130 days old. openFDA recalls were last updated by the FDA on 28 October. DailyMed and RxNorm were pulled on 5 November. Each source has its own clock, and the screen shows each one." Never say 'live' or 'real time'. Card 3 is a sliding 30-day window: 26 on 5 November, a different number on another day. Say so before anyone asks. Our pipeline refreshes only when a build runs: there is no schedule yet (another month).

Speed: what was measured

Machine timing (`app.timing_s`; browser script, time until the expected text is on screen, View mode, 2026-11-05): Morning brief loads in 11.5 s; page switches take 0.8 s (what changed), 1.1 s (recalls) and 2.4 s (signals with the labels panel). Human reading time is not included. These are the **only measured speed numbers**. 'Under two minutes' for her three questions is a design target, timed by a person with a stopwatch in rehearsal (checklist 10); until then say 'designed for under two minutes, not yet timed with a person'. The first load is why tab 1 is warmed up before the demo OBSERVED.

03

Part A: as the safety officer (0:00 to 7:30)

The app is the Workshop module **MSSM Safety Officer** (v0.9.0, header MSSM). It has four tabs: Morning brief, 1 · Not in label, 2 · Recalls, 3 · What changed. Deck slides 3 to 5 cover this part; the speaking script's **[CLICK]** cues match the clicks below.

A0 · The officer's problem (0:00–1:30, 90 s)

What	Detail
Click	Deck: slides 1 and 2
Say	"Every morning she has three questions: which drugs have side effects being reported that the label does not mention, which have an active recall, and what changed since she last looked. One rule: she trusts only a flag she can trace to its source. 13 drugs, 8 opioids and 5 GLP-1s, public data only, no patient data."
Proves	Communication

A1 · Morning brief (1:30–3:00, 90 s)

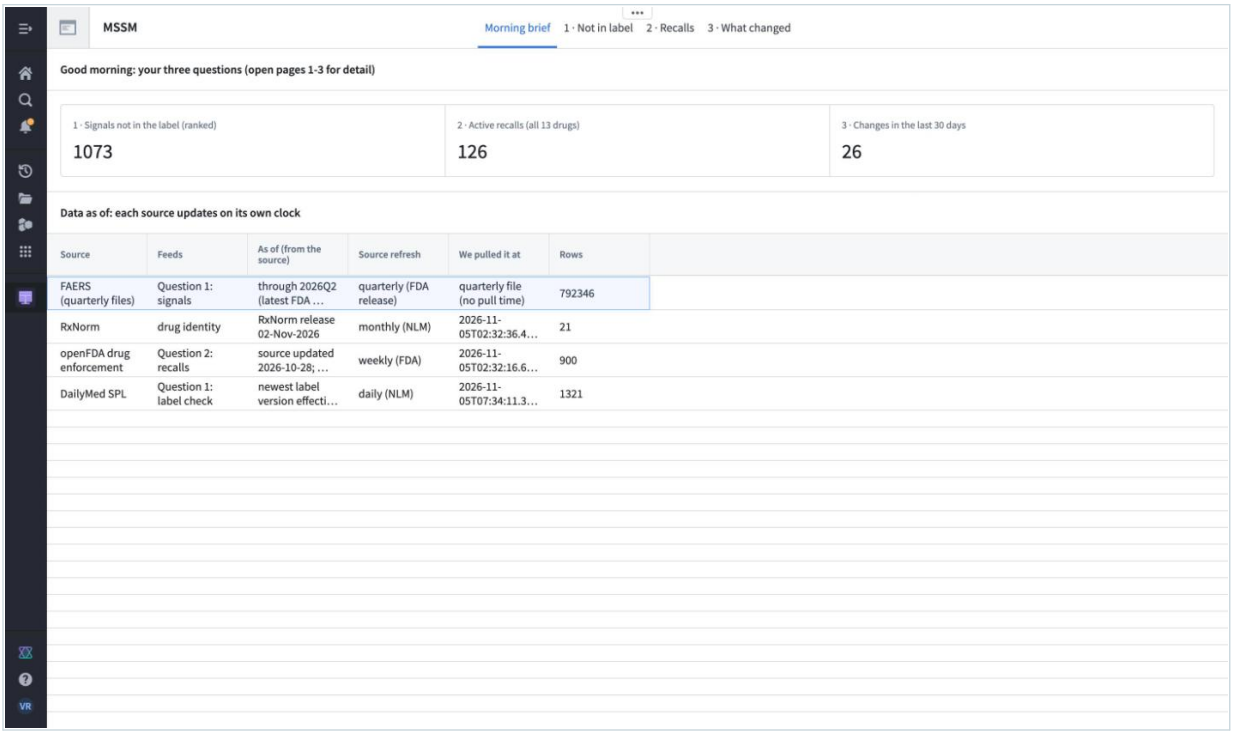


Figure 2. Morning brief: three numbers and the as-of date and pull time of each source.
Source: docs/11-handin/screens/app-0-morning-brief.jpg

What	Detail
Click	SWITCH TO APP: Cmd+1 (Workshop). Tab **Morning brief** is open and warmed up
On screen	Cards: 1 · Signals not in the label (ranked) **1073**; 2 · Active recalls **126**; 3 · Changes in the last 30 days **26**. Below, 'Data as of' per source. Column *We pulled it at*: FAERS = 'quarterly file (no pull time)'; RxNorm, openFDA and DailyMed = the pull time on 2026-11-05
Say	"One number per question. 1,073 is her review queue, not 1,073 findings. 126 recalls are active, meaning status Ongoing. 26 things changed in the last 30 days. Below is the trust part: how fresh each source is."
Proves	Application (three questions, one screen); trust (as-of per source)
Fallback	Cards spin over 10 s: backup window, app-0, keep talking
If asked	Why no pull time for FAERS? It arrives as quarterly files, so the screen says 'quarterly file (no pull time)' on purpose. Fast enough? The only measured numbers are machine timings (11.5 s first load); 'under two minutes' is a target until a person times it

A2 · The noise first, then one drug (3:00–3:40, 40 s)

What	Detail
Click	Tab **1 · Not in label**. **Before filtering**, point at row 5, **Morphine · Ewing's sarcoma**. Then pill **Ingredient Name**, tick **semaglutide**
On screen	Unfiltered: each drug's rank 1, including Morphine · Ewing's sarcoma and Liraglutide · Diabetic metabolic decompensation. After the filter, **Semaglutide · Optic ischaemic neuropathy** is rank 1
Say	"Look at morphine's rank 1: Ewing's sarcoma, a bone cancer. That is almost surely about the patients, not the drug. This is why 1,073 is a review queue, not a list of findings: she dismisses that in one click. Liraglutide's rank 1 is the disease it treats (DQ06). Now one drug: semaglutide, sold as Ozempic and Wegovy. Number one is NAION: sudden vision loss from poor blood flow to the optic nerve."
Proves	Application (one drug; the noise is named, not hidden)
Fallback	The A3 screenshot (unfiltered: morphine row 5, NAION row 1)
If asked	Why rank per drug? She reviews drug by drug, and the size of the statistic depends on how many reports a drug has. One overall ranking would let the busiest drug fill the top

A3 · The evidence: why NAION is flagged (3:40–4:30, 50 s)

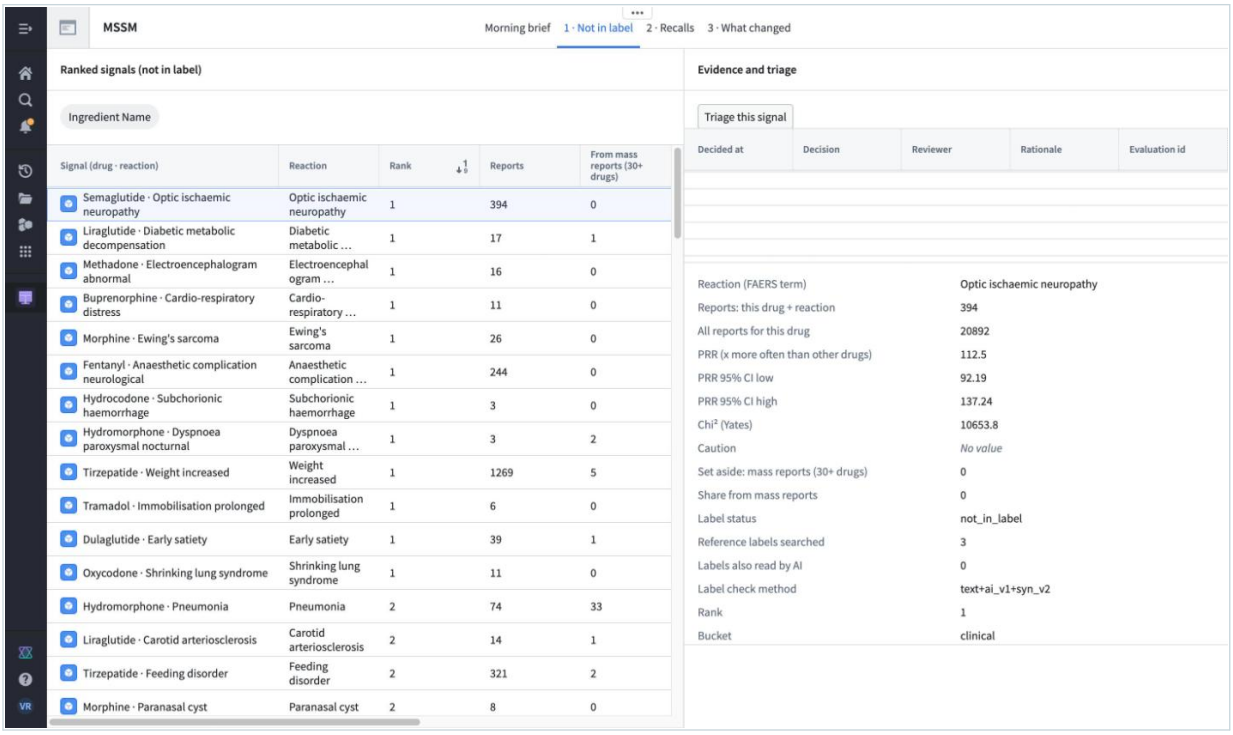


Figure 3. Page 1: ranked signals with the NAION evidence open.
Source: docs/11-handin/screens/app-1-not-in-label-naion.jpg

What	Detail
Click	The NAION row. The right panel 'Evidence and triage' fills
On screen	Reports 394 · all reports for the drug 20892 · PRR 112.5 · 95% CI 92.19 to 137.24 · Chi² 10653.8 · mass reports 0 · label status not_in_label · reference labels searched 3 · labels also read by AI 0 · method text+ai_v1+syn_v2 · bucket clinical. The decisions table is empty
Say	"394 reports name semaglutide and NAION, out of 20,892 semaglutide reports. NAION is about 112 times more common among semaglutide's reports than among other drugs' reports: a reporting pattern, not 112 times the risk. The interval runs from 92 to 137, so it is not a small-numbers fluke, but news coverage can still inflate it (DQ04). And NAION is not new: EMA's safety committee concluded in June 2025 that it is a very rare side effect of semaglutide. We lead with it because a monitor that missed it would be broken. Ours ranks it first."
Point	Do not read out the engineering fields. At *Label check method* text+ai_v1+syn_v2 say: "That code names the rule version, not which checks ran. The AI has not read these labels: look at *Labels also read by AI: 0*."
Proves	Application (trace on screen)
Fallback	This screenshot
If asked	Caused by semaglutide? We do not claim that: 'reported, not on the US label' (A003). Isn't NAION already known? Yes: EMA's PRAC, 6 June 2025; WHO alert, 27 June 2025; the 3 US label versions we search do not mention it (DECISIONS learning 2026-11-05). A known positive that the pipeline finds at rank 1 from FAERS alone; whether it would have flagged it earlier is unproven (backtest cut, D031)

A4 · Reference labels searched, then DailyMed itself (4:30–5:20, 50 s)

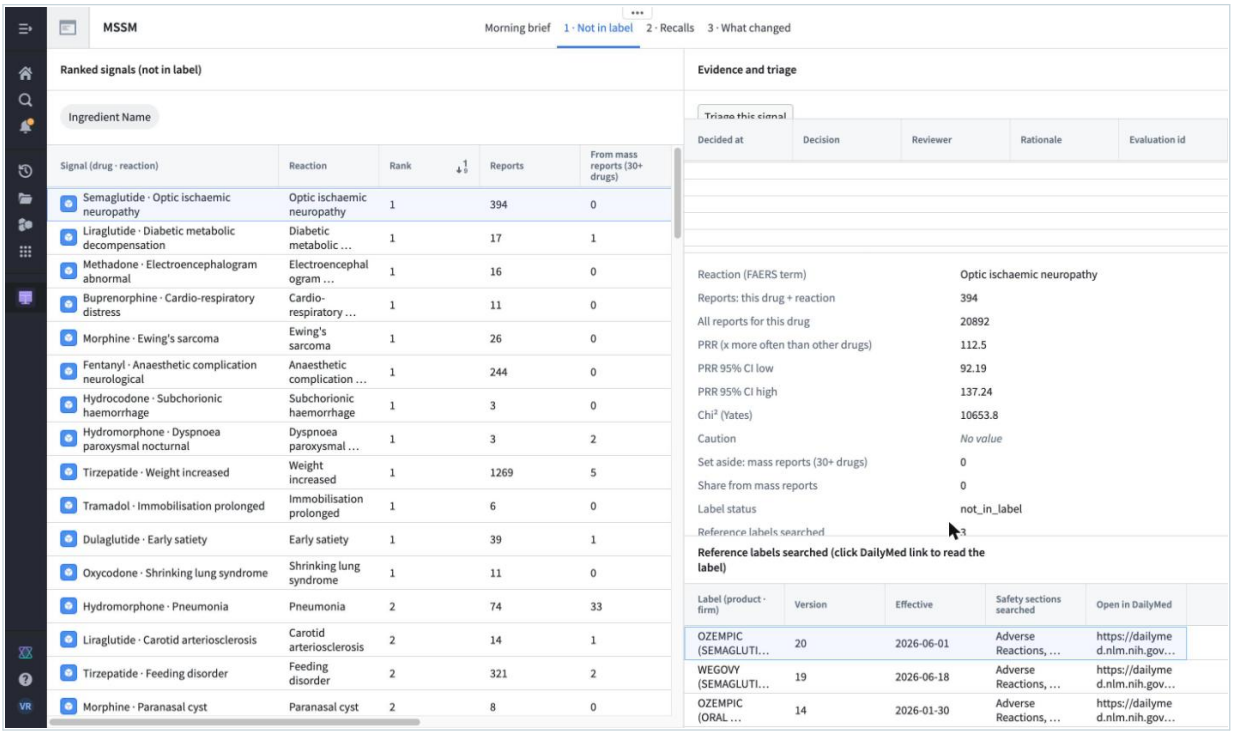


Figure 4. Page 1: the reference labels searched for NAION, by version, with DailyMed links.
Source: docs/11-handin/screens/app-1c-labels-searched-naion.jpg

What	Detail
Click	Scroll the right panel to **Reference labels searched**. **Mandatory:** click the **Ozempic v20** DailyMed link; Cmd+F `optic` (or `ischemic optic`, per checklist 8); show no optic neuropathy; Cmd+W back to the app
On screen	3 rows: **Ozempic v20** (2026-06-01), **Wegovy v19** (2026-06-18), **oral Ozempic v14** (2026-01-30); safety sections searched; a DailyMed link each
Say	"These are the three labels we searched, by exact version. Our exact-text check and reviewed synonym table find NAION in none of them; the AI has not read them. This is Ozempic's label on DailyMed itself. I search for 'optic': no optic neuropathy. She can repeat this check in ten seconds. What she cannot do yet is open the 394 reports from this screen: we pull their case IDs by query today, and that is the first thing we would add before anyone takes this to P&T."
Proves	Application (every flag traces to its source); AI reliability (says what the AI did not do)
Fallback	This screenshot; the DailyMed search screenshot from rehearsal (checklist 8)
If asked	How does the panel find them? Signal → Drug → Products → Reference Label; no dedicated link type yet. Check order? In code: (1) an AI term with a code-found quote, (2) the exact term or US spelling in a safety section, (3) a reviewed synonym. Same outcome in any order; only the recorded method differs (`ontology_signals.py` l.163–176). For NAION only (2) and (3) ran. Really missing? None of the 36 safety sections in any semaglutide label in DailyMed mention it (A014)

A5 · Triage this signal: the Action form (5:20–6:00, 40 s)

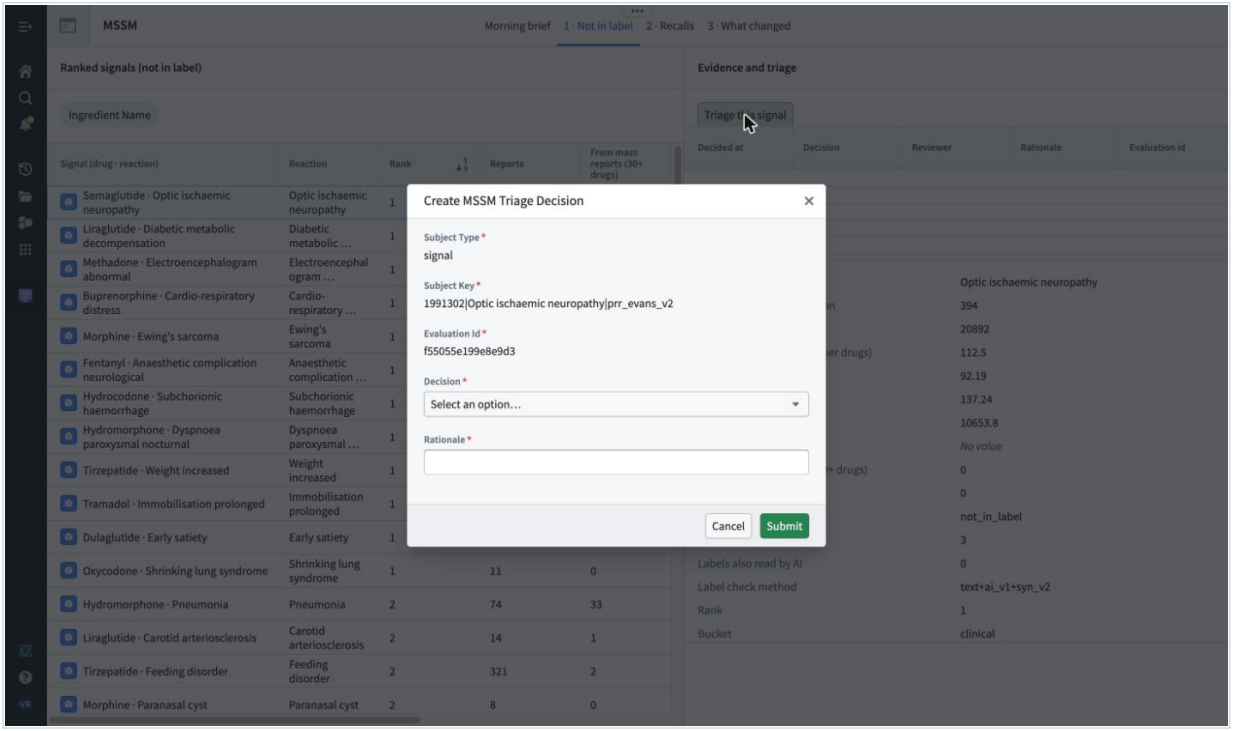


Figure 5. The triage form: subject and evaluation id locked (captured and cancelled, nothing saved).
Source: docs/11-handin/screens/app-1b-triage-action-form.jpg

What	Detail
Click	**Triage this signal**; point at the locked fields; **Cancel** (default). Only if the owner chose Submit: Decision **reviewing**, a one-line rationale, **Submit**; the row appears above the record
On screen	Form Create MSSM Triage Decision. Subject Type, Subject Key and Evaluation Id f55055e199e8e9d3 are filled and locked; Decision (reviewing, dismissed, escalated, monitoring) and Rationale are hers
Say	"This is a Foundry Action. The signal and its evaluation id are filled in and locked; she picks a decision and writes why. Who and when are set by the system. Submitting creates a new Triage Decision object; it never edits an old one, and it survives every rebuild. Today only the project owner can submit, so I will cancel."
Proves	Application (a triage button that writes back); trust (audit)
Fallback	Form not open within 10 s: this screenshot, same words
If asked	Evidence changes later? The decision keeps the id it was made on and stays linked to the signal (the link uses the signal key). The signal gets a new id. The screen does not flag the mismatch yet: she compares the id in the decisions table with the id the form shows. A bucket change does not change the id (ripple table)

A6 · Recalls, traced to the FDA's own record (6:00–6:50, 50 s)

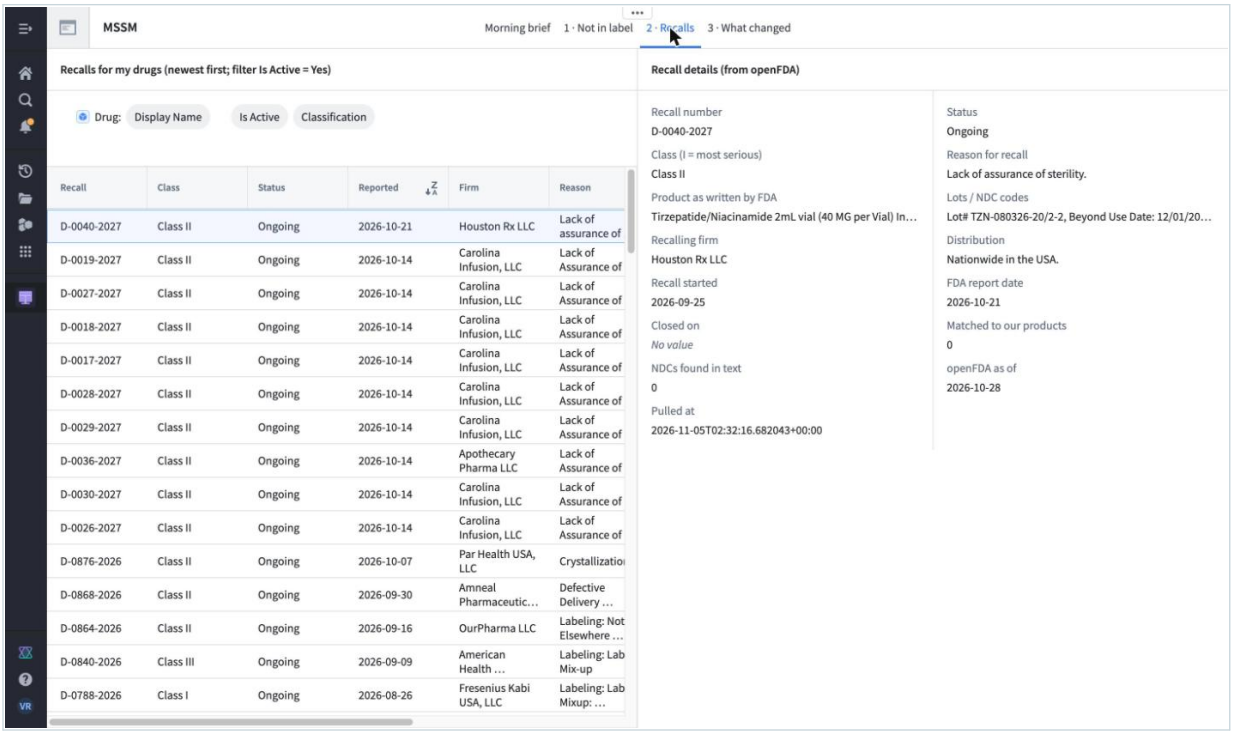


Figure 6. Page 2: recalls for the formulary, newest first, with the openFDA detail.
Source: docs/11-handin/screens/app-2-recalls.jpg

What	Detail
Click	Tab **2 · Recalls**. Is Active = Yes should be preset (deck script); if not, click the pill. Click the top row **D-0040-2027**; **Cmd+9** (the openFDA record, pre-opened): same number, same lot; **Cmd+1** back; click the **Class** column so Class I comes first
On screen	Recalls newest first. Detail: Class II · Ongoing · tirzepatide/niacinamide 2 mL vial · Houston Rx LLC · lack of assurance of sterility · reported 2026-10-21 · **matched to our products 0**
Say	"126 open FDA recalls, status Ongoing, name one of our 13 ingredients. The newest is a compounded tirzepatide vial, reported 21 October. It matches none of the 658 FDA-listed products we track: we have no inventory data, so if she buys compounded GLP-1s, she checks her purchasing records. Here is the FDA's own record of it: same number, same lot. Sorted by class, the Class I recalls come first." (Say the Class I count only if checklist 9 measured it)
Proves	Application (question 2); Data judgment (honest matching); trust (FDA record)
Fallback	This screenshot (D-0788-2026, Class I, lower in the list). openFDA tab fails: skip it
If asked	Which touch my shelves? We cannot know without purchasing data; 'matched' means the 658 FDA-listed products, via NDC codes in the recall text. Class I count and D-0788-2026's match: not measured unless checklist 9 was done. Active = openFDA status Ongoing (`clean_recalls.py` l.69); Completed and Terminated are not active. Recall to drug: the ingredient named in the recall text. Recall to product: NDC codes in the recall's own text, high or medium confidence

A7 · What changed (6:50–7:30, 40 s)

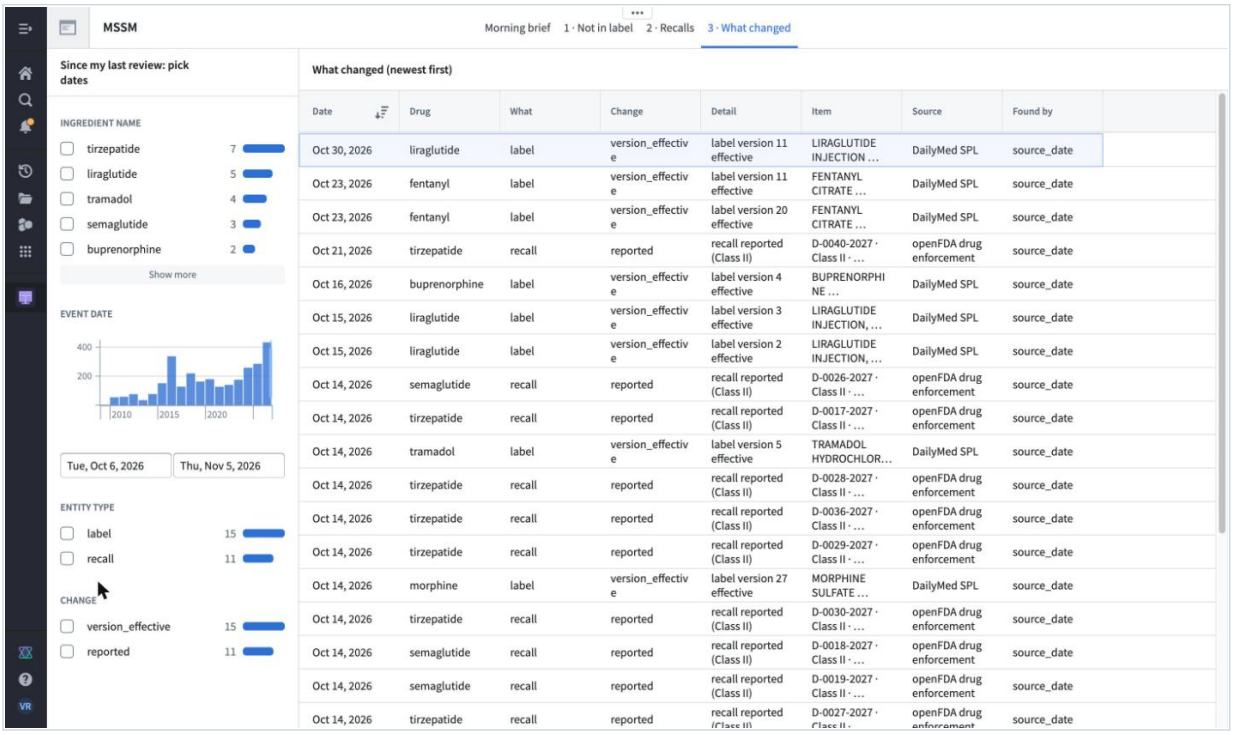


Figure 7. Page 3 with the Past month preset: 26 events, the same as card 3.
Source: docs/11-handin/screens/app-3b-what-changed-past-month.jpg

What	Detail
Click	Tab **3 · What changed**; Start date box, preset **Past month** (or the date of her last review)
On screen	Oct 6 to Nov 5: label versions effective (liraglutide v11, 30 Oct), recall D-0040-2027 reported (21 Oct). Entity type: label 15, recall 11 = 26, the same as card 3
Say	"Every change is an event, dated by the source itself. 'Since my last review' is simply this start date: the past month gives 26 events, the same as the morning brief. A label row tells her a new version took effect, not what changed in it; on DailyMed she opens Recent Major Changes. And new signals arrive only with each FAERS quarter, about 130 days behind. So that is her morning today: the sources publish daily (DailyMed) and weekly (openFDA); our pipeline refreshes when a build runs, with no schedule yet; signals move once a quarter."
Proves	Application (question 3)
Fallback	This screenshot (or app-3, unfiltered)
If asked	False alarms? A rebuild with no new data added 0 events. No signal changes yet: they come only from comparing builds, so they appear after the next FAERS quarter or synonym edit. The start date does not fill itself from her last decision yet (D045, proposed)

04

Part H: under the hood (7:30 to 20:00)

One stop per deck slide (6 to 16), plus the live switches the speaking script cues. Switches marked optional are the first cuts when late.

H1 · Architecture in one picture (7:30–8:30, 60 s)

What	Detail
Click	BACK TO DECK: slide 6, diagram 38 (the one-page architecture picture). Detailed fallback if asked: diagram 30 v2 (`docs/diagrams/png/30-handin-data-flow-v2.png`, every dataset, O3 measured)
Say	"Four public sources. Raw keeps exactly what each source sent. Clean checks and de-duplicates it, and resolves drug identity once, in one dataset: clean_drug_identity. Curated holds the statistics, signals, change events and monitoring. The Ontology sits on top, and the app reads only the Ontology."
Proves	Architecture

H2 · Data Lineage, live (8:30–9:30, 60 s)

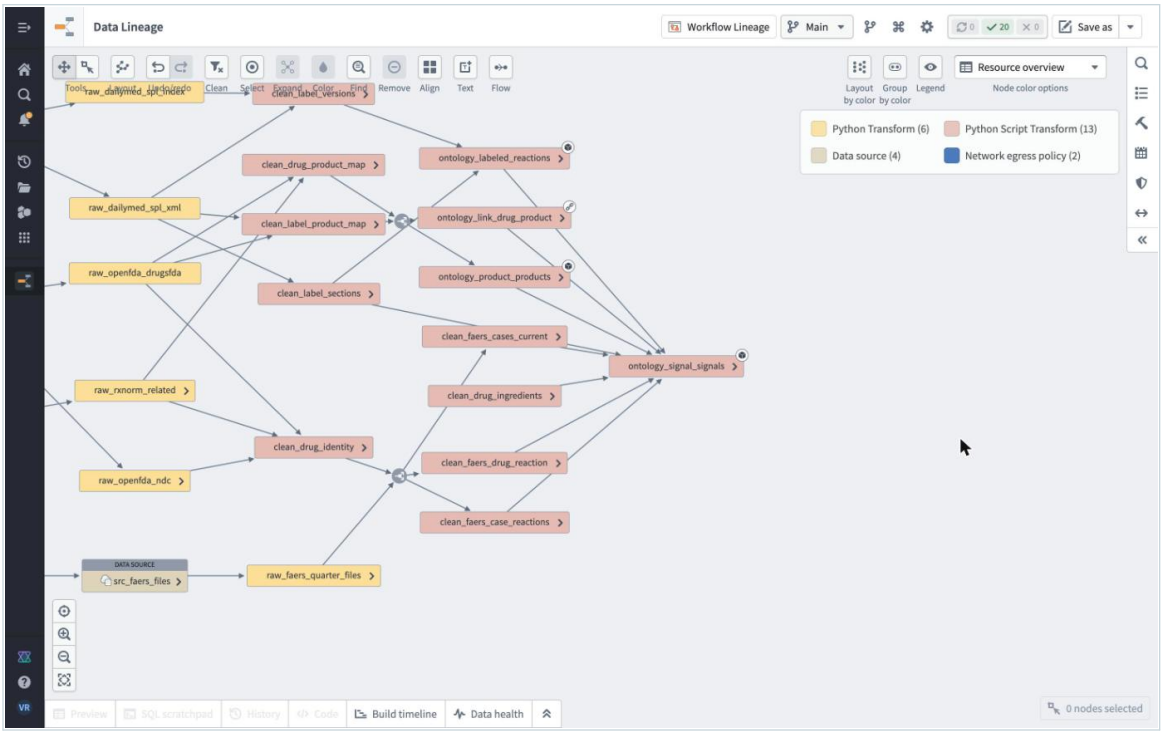


Figure 8. Data Lineage: sources, raw, clean and curated, ending in ontology_signal_signals.
Source: docs/11-handin/screens/hood-data-lineage-signals.jpg

What	Detail
Click	SWITCH: Cmd+2 (graph already laid out)
On screen	Yellow raw transforms (raw_dailymed_spl_xml, raw_openfda_drugsfda, raw_rxnorm_related, raw_openfda_ndc, raw_faers_quarter_files from source src_faers_files), pink clean transforms, ending in ontology_signal_signals
Say	"This is Foundry's own graph, not a drawing. Follow RxNorm into clean_drug_identity. Everything after it, FAERS, labels and products, joins on one ingredient ID. That is 'drug identity resolved once, in the right place'."
Proves	Architecture
Fallback	This screenshot (deck backup B12), or diagram 30 v2
If asked	Why polars, not Spark? Lightweight transforms took about 1 min against about 4 min for Spark (learning 2026-11-04), and the same code runs on a laptop, which is how we test it (A014)

H3 · The Ontology in one picture (9:30–10:15, 45 s)

What	Detail
Click	BACK TO DECK: slide 7, diagram 31 (ontology)
Say	"Ten object types: Drug, Product, Label, Recall, Labeled Reaction, AE Stat, Signal, Change Event, Source Status, Triage Decision. Only Triage Decision is written, and only through one create-only Action. One mistake is on the picture on purpose: a duplicate Signal-to-Drug link added by accident. Nothing uses it; it is left for the owner to delete."
Proves	Architecture

H4 · Ontology Manager (10:15–10:45, 30 s)

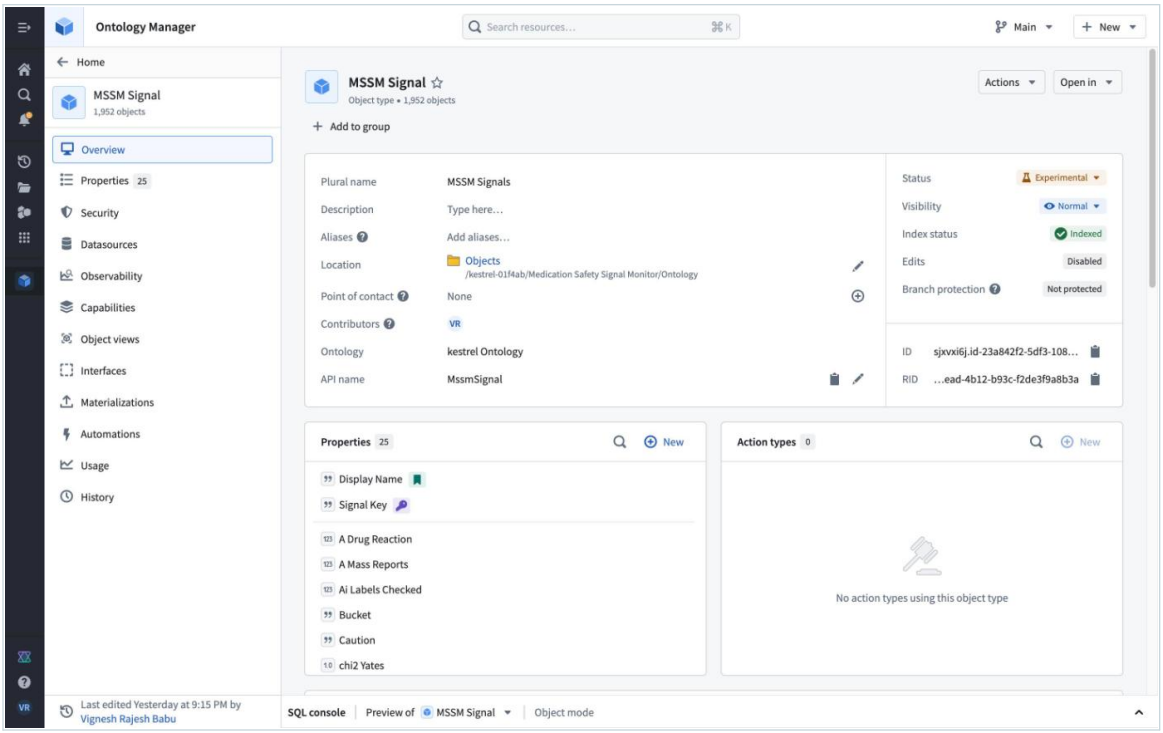


Figure 9. Ontology Manager: the MSSM Signal object type.
Source: docs/11-handin/screens/hood-ontology-manager-signal.jpg

What	Detail
Click	SWITCH: Cmd+3
On screen	MSSM Signal · 1,952 objects · 25 properties · Edits disabled · Experimental
Say	"Each object type is backed by one curated dataset. Signal has 1,952 objects. Edits are off: signals come only from the pipeline. The Action lives on Triage Decision, which points at the signal."
Proves	Architecture
Fallback	This screenshot

H5 · Object Explorer: follow one flag to its source (10:45–11:30, 45 s)

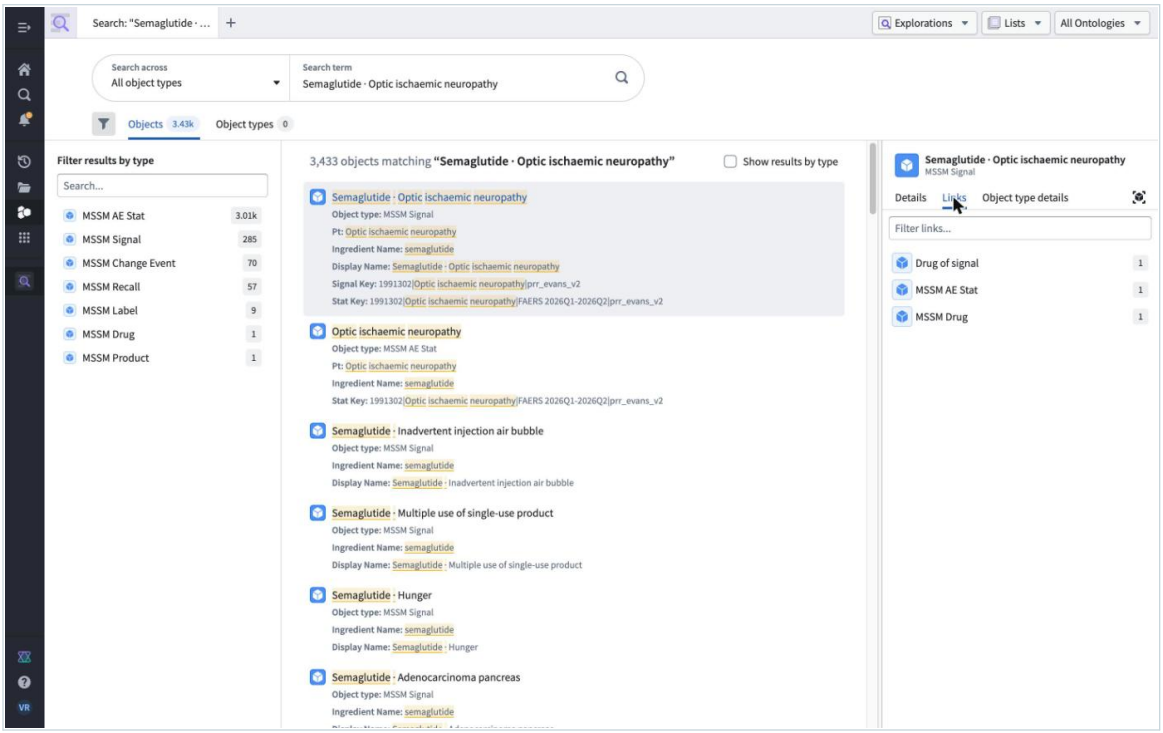


Figure 10. Object Explorer: the NAION signal and its links.
Source: docs/11-handin/screens/hood-object-explorer-naion-links.jpg

What	Detail
Click	SWITCH: Cmd+4 (search done, NAION selected, Links tab); MSSM AE Stat; back; MSSM Drug, MSSM Products, Reference Label (DailyMed link)
On screen	Links: Drug of signal 1 (the duplicate) · MSSM AE Stat 1 · MSSM Drug 1. No Triage Decisions link, because the form was cancelled (with Submit: MSSM Triage Decisions 1)
Say	"The trust rule as data. From the signal to its statistics, the 2-by-2 table of counts. From the drug to its products and the reference label, with the DailyMed link: the same path the page 1 panel walks."
Proves	Application (trace: every flag traces to its source)
Fallback	This screenshot (deck backup B4); the baseline-gate trace (5 of 5 random flags re-counted by hand)
If asked	FAERS case IDs: SQL on clean_faers_drug_reaction (the baseline gate lists examples); the screen does not link to case rows yet

H6 · One FAERS report to a ranked signal (11:30–12:30, 60 s)

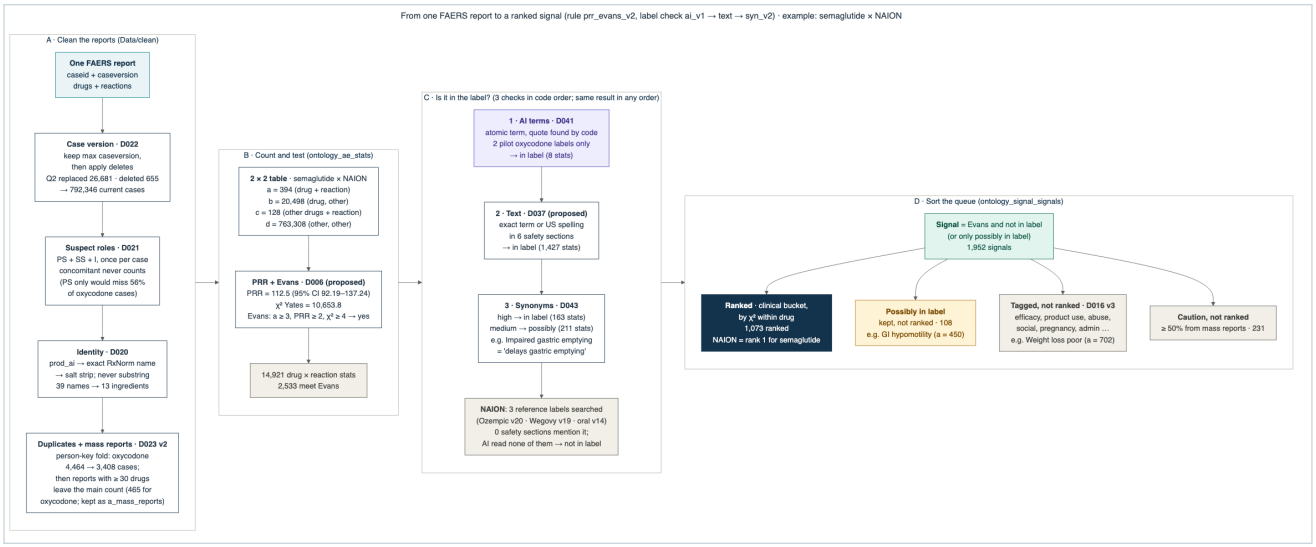


Figure 11. Diagram 32 v2: from one FAERS report to a ranked signal.
Source: docs/diagrams/png/32-handin-signal-pipeline-v2.png

What	Detail
Click	BACK TO DECK: slide 8, diagram 32 v2 (signal pipeline, label checks in code order)
Say	"A: keep only the latest version of each case and apply deletes to all history; appending quarters gave 8.7% phantom rows. Count only suspect drugs. Match names exactly through RxNorm; substring matching added 50% false morphine rows. Set aside reports listing 30 or more drugs. B: the 2-by-2 table, PRR and the Evans rule. C: in the label? In code: an AI term with a quote, then the exact text, then the reviewed synonym table. D: only clinical reactions are ranked; the rest is tagged."
Proves	Data judgment
Sources	DQ36 and D022 (versions, deletes), D021 (roles), DQ34 and D020 (identity), D023 v2 (mass reports), D006 (Evans, proposed), D043 (synonyms), D016 v3 (buckets)

H7 · The signal code (optional) (12:30–12:50, 20 s)

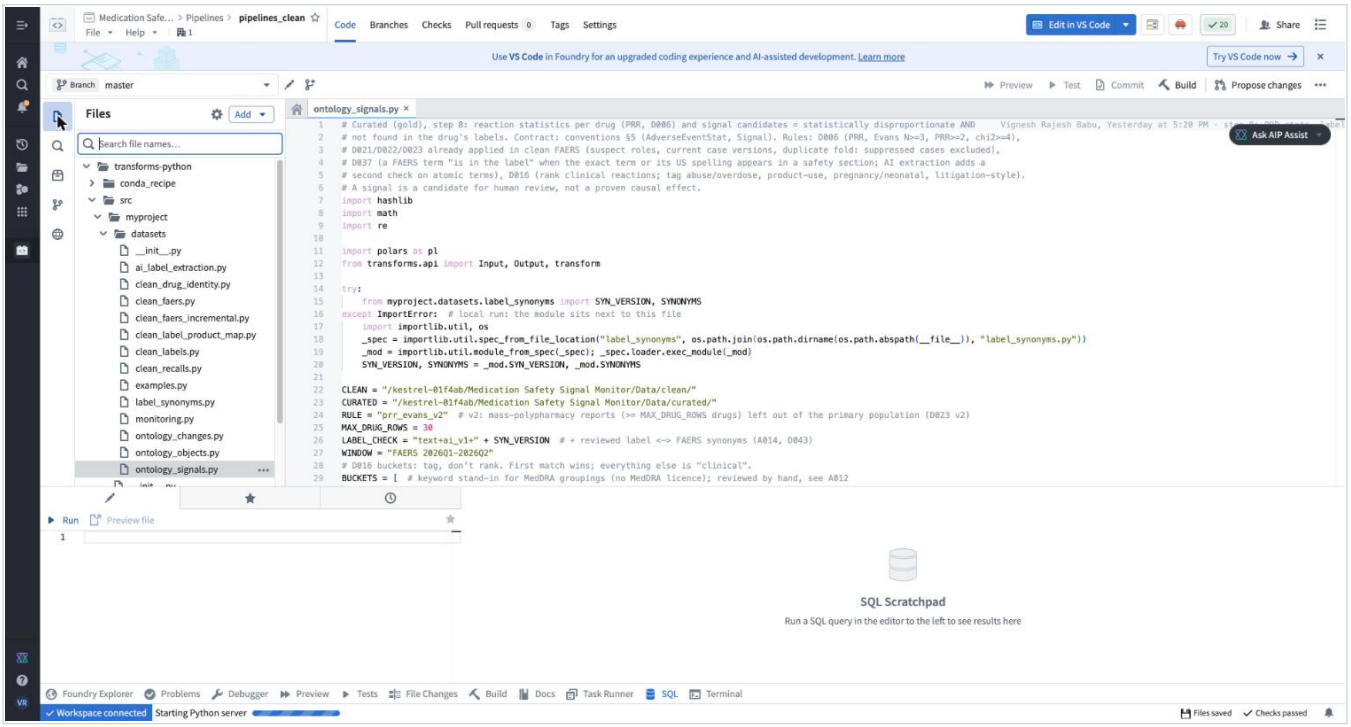


Figure 12. Code Repository pipelines_clean: ontology_signals.py and the file tree.
Source: docs/11-handin/screens/hood-code-repo-ontology-signals.jpg

What	Detail
Click	Optional SWITCH: Cmd+5 (ontology_signals.py, lines 22 to 29). First cut if late
On screen	RULE = "prr_evans_v2" · MAX_DRUG_ROWS = 30 · LABEL_CHECK = "text+ai_v1+" + SYN_VERSION · WINDOW = "FAERS 2026Q1-2026Q2" · BUCKETS: keyword stand-in for MedDRA groupings (no MedDRA licence)
Say	"Each rule is a named constant with a decision number. The buckets are keyword lists because we have no MedDRA licence, and the code says so."
Proves	Architecture (another engineer can extend it)
Fallback	This screenshot

H8 · A review queue, not findings (12:50–13:30, 40 s)

What	Detail
Click	BACK TO DECK: slide 9
Say	"14,921 drug-reaction statistics, 1,952 signals, 1,073 ranked. The queue after each label-check or bucket change: 1,355 at step 9, our first full build, then 1,210, 1,115 and 1,073. 108 are 'possibly in label', kept but never ranked. The 258-row synonym table was drafted by Claude; no pharmacist has reviewed it yet."
Proves	Data judgment

H9 · AI reliability (13:30–14:45, 75 s)

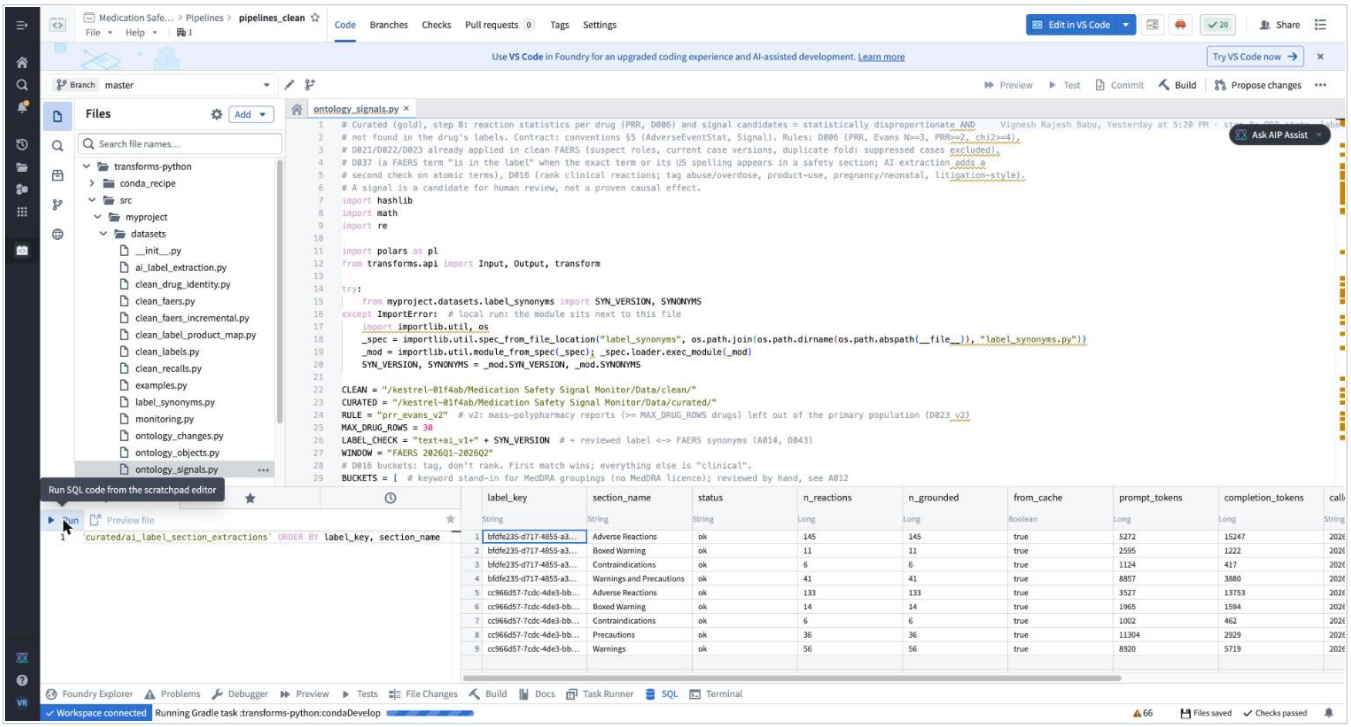


Figure 13. SQL on ai_label_section_extractions: every section grounded and served from cache.
Source: docs/11-handin/screens/hood-ai-extractions-cache-sql.jpg

What	Detail
Click	Slide 10. Optional SWITCH: Cmd+6, ai_label_extraction.py lines 16 to 25, then the SQL panel (AI cache query already run)
On screen	MODEL_ID "claude-sonnet-5.5" · PROMPT_VERSION "p2" · PARSER_VERSION "x1" · PILOT_SETIDS (2 oxycodone labels). SQL: 9 rows, n_reactions = n_grounded, from_cache true
Say	"Claude Sonnet 5.5 read 9 safety sections of 2 oxycodone pilot labels and returned each reaction with an exact quote. Code finds every quote word for word: 448 of 448. On a small answer key, 35 items from the Boxed Warning and Contraindications of those 2 labels, written by Claude before the run, it found 35 of 35 answer-key items: F1 0.97 against a target of 0.85. No pharmacist has checked the key, and the long sections are not scored. The AI has not read semaglutide labels, and today it does not change the signal list. On the last rebuild: 0 model calls."
Proves	AI reliability; Optimization (O1)
Fallback	This screenshot and hood-code-ai-label-extraction (also deck/assets/ai-cache-sql-crop.png)

H10 · Monitoring (14:45–15:30, 45 s)

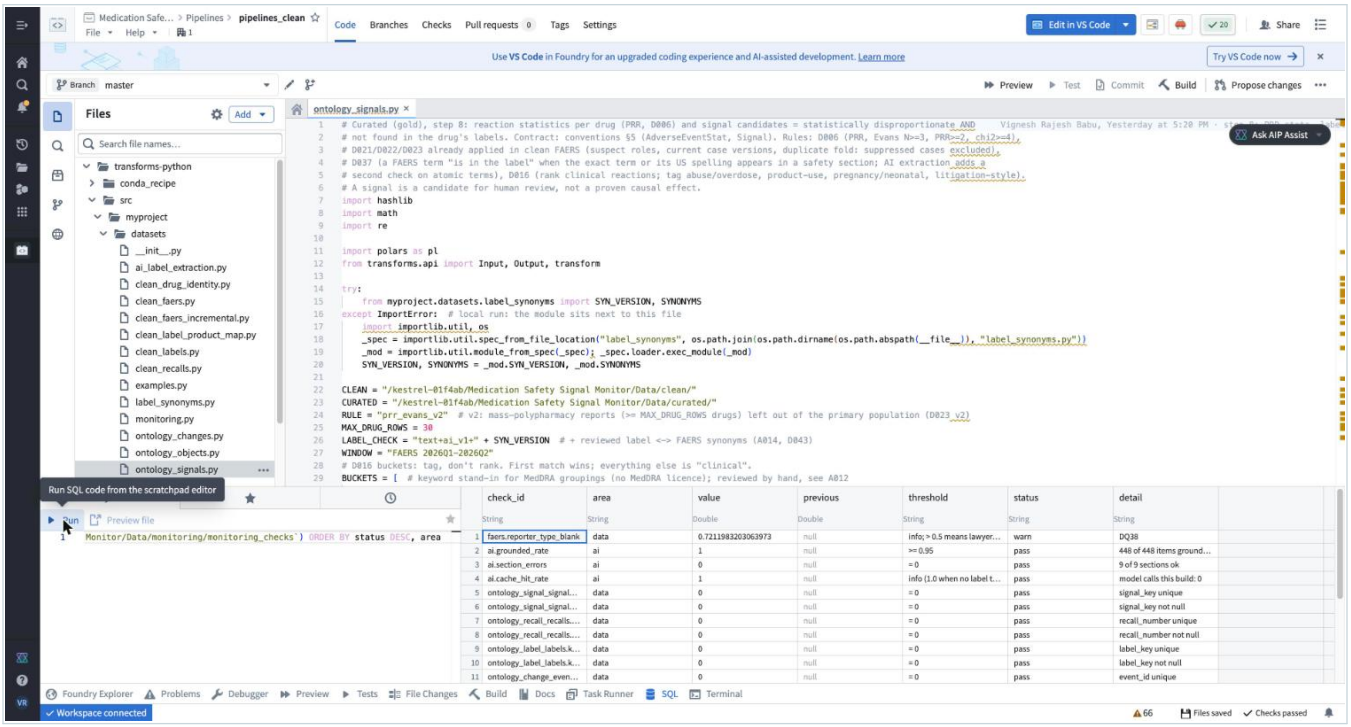


Figure 14. SQL on monitoring_checks: the latest run, one real warning (DQ38).
Source: docs/11-handin/screens/hood-monitoring-checks-sql.jpg

What	Detail
Click	Slide 11. Optional SWITCH: Cmd+5, SQL panel (monitoring query already run), or Cmd+7, Health tab
On screen	faers.reporter_type_blank **warn**; ai.grounded_rate 1 pass; ai.section_errors 0 pass; ai.cache_hit_rate 1; key checks pass. Health tab: Build status and Primary key Passed
Say	"Every build appends 21 checks: keys, row counts, freshness, AI grounding, cache hits; plus 4 Foundry Data Health checks. It caught two real shifts: reporter type blank went from 34% to 72%, and recalls for our drugs from 3 to 10 in four weeks. The drift is a replay from source dates, not weeks of live builds."
Proves	AI reliability (stretch goal: notice if it gets worse); Data judgment
Fallback	This screenshot; hood-data-health-signals; hood-monitoring-drift-sql

H11 · Optimization log (15:30–16:45, 75 s)

What	Detail
Click	Slide 12
Say	"O1, the AI cache: 9 model calls and 70,776 tokens became zero on a rebuild. Helped. O2, incremental FAERS: 88 seconds against 32. Did not help. O3, reusing DailyMed XML we already hold: 3 minutes 23 to 1 minute 23, and 1,321 XML downloads to 2; the index pages are still pulled every build. Helped. O3a, parsing fewer columns: 12.3-13.0 s to 12.2 s, locally; reverted. Partitioning: none in the pilot (D048, proposed); we revisit at 8 FAERS quarters or a 5-minute build. Each timing is one run."
Proves	Optimization (measured, failures included)

H12 · Incremental FAERS (16:45–17:30, 45 s)

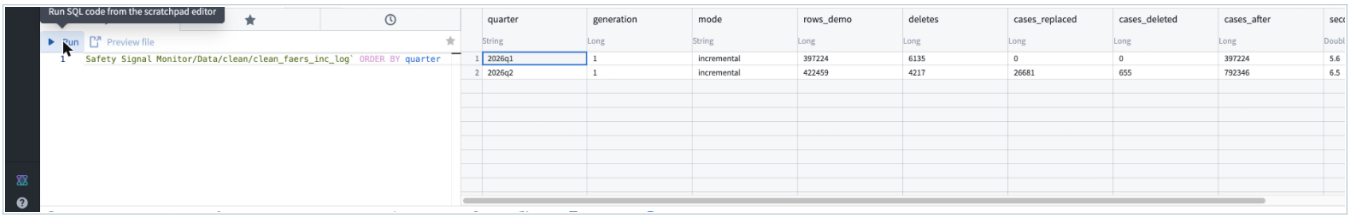


Figure 15. SQL on clean_faers_inc_log: one row per quarter, with cases replaced and deleted.
Source: docs/11-handin/screens/hood-faers-inc-log-sql.png

What	Detail
Click	Slide 13 (diagram 33 v2). Optional SWITCH: Cmd+8, clean_faers_inc_log preview
On screen	2026q1: 397,224 cases; 2026q2: 26,681 replaced, 655 deleted, 792,346 after
Say	"The incremental path reads only the new quarter into a kept state: three state datasets and one quarter log. Its output equals the full rebuild row for row. But at two quarters it is slower: 62 s for the state plus 26 s for the outputs, against 32 s. So signals keep reading the full rebuild (D044), and we re-measure at four or more quarters."
Proves	Optimization (incremental against full, measured)
Fallback	This screenshot

H13 · Data judgment and what we chose not to do (17:30–19:00, 90 s)

What	Detail
Click	Slide 14. Optional SWITCH: Cmd+5, SQL panel, point at faers.reporter_type_blank **warn**
Say	"Four messes that would have changed her answers. FAERS updates and deletes old cases: Q2 replaced 26,681 Q1 cases and deleted 655. Mass reports listing dozens of drugs: out of the main count, kept as evidence. Reporter type went from 34% to 72% blank: we show 'unknown'. DailyMed sometimes answers 'success' with the word 'None' (2 of 1,321, DQ42): kept, marked, retried. Noise we left: indication terms, like liraglutide's top signal 'Diabetic metabolic decompensation' (DQ06), and stimulated reporting after news or lawsuits (DQ04); the pharmacist's triage decides. Not done: no MedDRA licence, no patient data."
Proves	Data judgment
Sources	DQ36, D022, D023 v2, DQ38, DQ42, DQ06, DQ04
Fallback	The H10 screenshot (also deck/assets/monitoring-checks-sql-crop.png)

H14 · Another month (19:00–19:40, 40 s)

What	Detail
Click	Slide 15
Say	"The biggest gap is a named pharmacist. Today only Claude has checked the synonym bands, the answer key and the ranked signals. Claude prepares an evidence packet; the pharmacist decides; a create-only Action records who, when and on which evidence. Then schedules, live drift over three to four weeks, and a health page. Widening the AI to the semaglutide labels is the owner's next decision (D041 v2, not written yet)."
Proves	Communication

H15 · What to take away (19:40–20:00, 20 s)

What	Detail
Click	Slide 16
Say	"It answers her three questions, and every flag traces to its source: the label versions searched and the FDA recall record open in one click; the FAERS case reports are reachable by query, not yet from her screen. It is built to be extended, and we sped up only what we measured. And we say what is not proven yet, including that the AI has not read the hero drug's labels."
Proves	Communication

05

Plan B and hard questions

Rule from rehearsal: a page that takes more than 10 seconds to load in rehearsal is shown as a screenshot in the demo. The backup window holds every screenshot in this guide.

#	Moment	What can go wrong	Plan B
1	1:30 Morning brief first load	Cards spin; session timed out	Warm all 4 tabs 5 minutes before. Over 10 s: the A1 screenshot, keep talking, rejoin the live app at A2
2	5:20 Triage form	Slow to open; Submit pressed by mistake	Default is Cancel. Slow: the A5 screenshot, same words. A Submit by mistake cannot be undone: say so plainly ('this is what create-only means') and move on
3	8:30 Data Lineage	Slow, or re-laid-out into a tangle	Lay out during setup; never press Layout. Messy: the H2 screenshot
4	12:30 to 17:30 Code and SQL	Repo still starting; SQL slow; accidental keystroke	Open 10 minutes early, pre-run the queries, never type in the editor (close the tab without committing if you do). Slow: stay on the slide
5	Any time: numbers differ from slides	Card 3 slides with the date; someone rebuilt raw	Say it first: '26 on 5 November; it is a sliding window'. If cards 1 or 2 moved: 'the data refreshed since our baseline; the slides are from 5 November'

The three rubric-check questions, prepared

A reviewer is likely to ask these (from `reviews/rubric-check.md`). Each answer is said honestly, and each has a place to click.

Question	Answer	Where to show it
Your AI read two labels and does not change the list. What is it doing, and how often is it wrong?	On purpose, the AI is the evidence layer today, not the decision layer. It has read 0 semaglutide labels, so NAION is not in label from the exact-text and synonym checks. Every AI quote is found word for word (448 of 448); on the 35-item Claude-written key (2 oxycodone labels, Boxed Warning + Contraindications, no pharmacist yet) it found 35 of 35 answer-key items, F1 0.97. Widening to semaglutide is a D041 v2 decision for the owner: first a pharmacist-scored sample of the long sections, then widen	Page 1 NAION: labels also read by AI 0. Tab 6: PILOT_SETIDS (line 22). Q&A SQL on ontology_labeled_reactions: term, grounding, quote. Deck slide 10
Liraglutide's top 'unlabeled' signal is the disease it treats.	Yes: 'Diabetic metabolic decompensation', an indication term (DQ06), left and named. D016 v3 tags some efficacy words, not this one. With another month: an indication-term list per drug; until then the pharmacist's triage dismisses it	Page 1 unfiltered, row 2 (A3 screenshot), or filter liraglutide. Tab 5: BUCKETS, efficacy line
How many of NAION's 394 reports came after news or from lawyers?	Not measured. Stimulated reporting (DQ04) is a known gap we left; triage judges it. A reporter split is weak anyway: reporter type is blank for 34% (Q1) and 72% (Q2) of reports (DQ38). With another month: NAION's reporter-type split, and a spike flag near known events	Tab 5 SQL: faers.reporter_type_blank warn; drift replay screenshot (reporter type blank x2.1). Deck slide 14

Change here, changes there (summary)

Judges often ask "what happens if …". The full table, checked against the code, is `docs/11-handin/demo/ripple-table.md`. Nothing is changed live; build times not in numbers.json are not measured, so do not promise them.

Change this here	This changes there	Evaluation id	Show
Synonym row (+ SYN_VERSION)	label status, ranked queue, build-dated change events	every signal	history: 1,355 to 1,073
Bucket regex	bucket, is_ranked, change events	unchanged	the code
New FAERS quarter (+ WINDOW)	counts, PRR, every AE Stat key, signals, freshness	yes	inc log
Drug added (5 formulary lists)	every layer; FAERS state replays all quarters	new signals	no
New DailyMed label version	one download; label check; one build-dated 'spl version' event	only if label counts change	page 3 history
Recall status change	card 2, page 2; one build-dated 'status' event	n/a	page 3 history
Triage decision	one Ontology object; nothing else; cannot be deleted	stores it	form, Cancel
Prompt or model version	cache key; all pilot sections re-sent	only if AI label count changes	code
Workshop variable	that card only; no build	n/a	definition only

What happens to an existing triage decision when the evaluation id changes

The decision is not edited and stays linked to its signal (the link uses the signal key). It keeps the old evaluation id, so it still shows on page 1 under the signal. Nothing on screen flags the mismatch, and the change feed does not watch the evaluation id. A bucket change keeps the id. If the signal leaves (now in the label), the decision stays in Object Explorer but no longer shows on page 1.

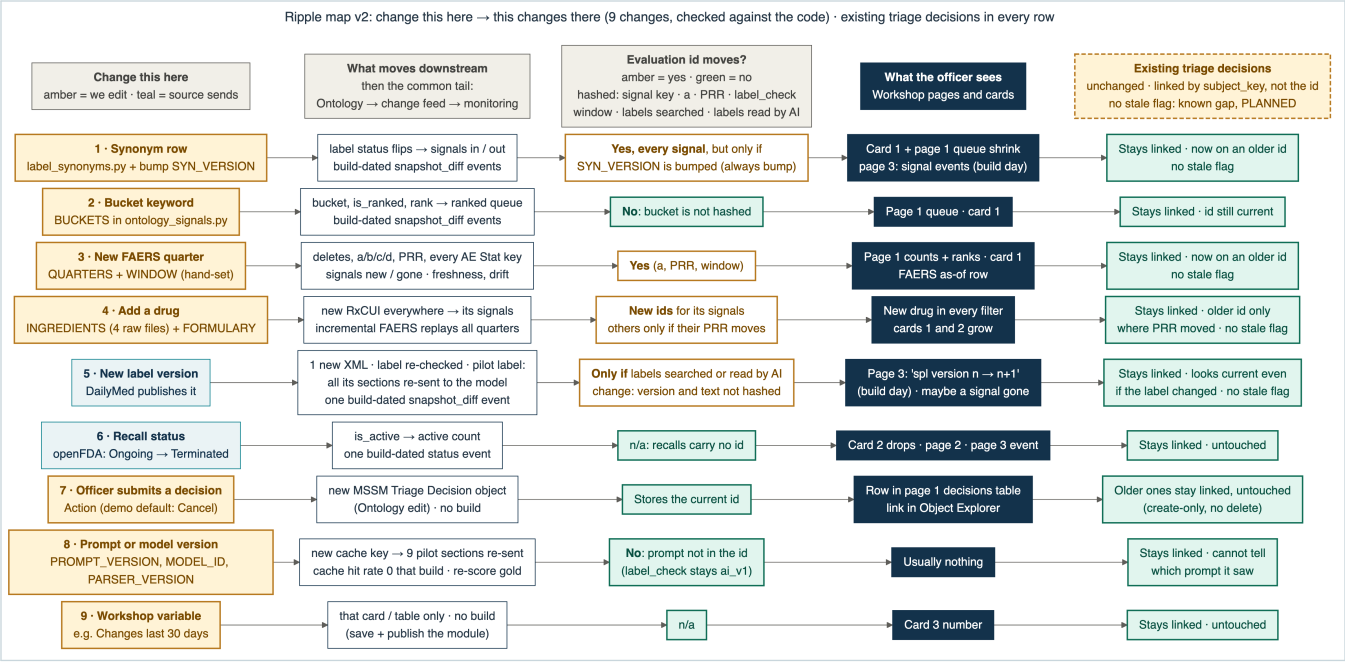


Figure 16. Ripple map (diagram 37 v2): change here, changes there.
Source: docs/diagrams/png/37-handin-ripple-v2.png

Gate: Demo ready PLANNED

Required evidence: Two rehearsal passes done (tech run, dress run) with measured seconds per stop; every live stop has a fallback screenshot open in the backup window; the owner's Cancel or Submit choice is written down.

Result: Guide, ripple table and rehearsal plan written and aligned with the speaking script. Machine timings measured (brief 11.5 s). Rehearsals and a human timed walk-through not yet run.

If it fails: Use screenshots for every stop slower than 10 s, and take the cuts in chapter 1.

Numbers used in this report

Every number in the table below is looked up from `docs/11-handin/numbers.json` at build time. Source: Foundry SQL (repo SQL panel) after the baseline-gate rebuild of the whole downstream chain: raw_dailymed_spl → clean_labels → clean_label_product_map → ontology_objects → ai_label_extraction → ontology_signals → ontology_changes → monitoring. Build durations from the Foundry Builds page. Query: docs/11-handin/baseline-gate.md. Captured: 2026-11-05T07:50Z. Figures quoted from other project records (decision log, data-issues register, optimization log) name that record where they appear.

Key	Value
ai.d015_target_f1	0.85
ai.gold_f1	0.97
ai.grounded	448
ai.model_calls_on_rebuild	0
ai.pilot_labels	2
ai.reactions	448
ai.sections	9

Key	Value
app.labels_searched_naion	3
app.pages	4
app.timing_s.morning_brief_load	11.5
app.timing_s.switch_recalls	1.1
app.timing_s.switch_signals_with_labels	2.4
app.timing_s.switch_what_changed	0.8
app.triage_decisions	2
app.workshop_version	v0.9.0
changes.last_30_days	26
changes.snapshot_diff_after_rebuild	0
faers.cases_current	792,346
faers.q1_cases	397,224
faers.q2_cases_deleted	655
faers.q2_cases_replaced	26,681
monitoring.checks_per_build	21
monitoring.data_health_checks	4
monitoring.faers_age_days	130
monitoring.recalls_last4w	10
monitoring.recalls_prev4w	3
monitoring.reporter_type_blank_q1	0.343
monitoring.reporter_type_blank_q2	0.721
optimization.O1.before_calls	9
optimization.O1.before_tokens	70,776
optimization.O2.full_rebuild_s	32
optimization.O2.incremental_add_q2_s	88
optimization.O2.outputs_s	26
optimization.O2.state_s	62
optimization.O3.after_downloads	2
optimization.O3.before_downloads	1,321
optimization.O3a.local_run_after_s	12.2
optimization.O3a.local_run_before_s	12.3-13.0

Key	Value
recalls.active	126
recalls.newest	D-0040-2027
recalls.newest_reported	2026-10-21
scope.drugs_formulary	13
scope.drugs_with_signals	12
scope.glp1	5
scope.opioids	8
signals.ae_stats	14,921
signals.naion.a	394
signals.naion.all_reports_drug	20,892
signals.naion.chi2_yates	10653.8
signals.naion.prr	112.5
signals.naion.prr_ci_high	137.24
signals.naion.prr_ci_low	92.19
signals.possibly_in_label	108
signals.ranked	1,073
signals.ranked_history.d016_v3	1,115
signals.ranked_history.step9	1,355
signals.ranked_history.syn_v1	1,210
signals.ranked_history.syn_v2	1,073
signals.total	1,952
synonyms.rows	258